

Imaging of Biological Systems with Fluorescent Technologies

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Fluorescence microscopy imaging has become one of the most useful techniques in biomedical research, used to assess the activity of individual cells as well as subcellular processes. The past 2 decades have seen tremendous advances in the rational design and development of probes to serve as reporters of the intracellular environment of cells. New fluorescent molecules have been engineered to follow signaling traffic, sense ionic and nonionic secondary messengers, and report various kinase activities. In particular, the advent of GFP cloning has provided a means of making specific proteins visible in living cells, allowing the study of molecular behavior and interactions with high spatial and temporal resolution. It is now possible to examine these events in intact tissue, enabling a more complete understanding of complex physiologic processes.
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The predominant focus of biomedical research in the post-genomic era is directed toward identifying the function of molecules at the cellular and subcellular levels in living systems, such that an integrated understanding of their function is possible. This goal demands both temporal and spatial analysis that is simply not possible using methods dependent on the extraction or disruption of cells in which the events of interest are occurring.

Fortunately, the “post-genomic” need to study molecular function has been paralleled by extraordinary progress in optical microscopy and computer technologies. Contemporary live cell microscopy can be used to examine the expression, functional role(s), and interactions of multiple molecules concurrently within the cellular environment. Also, new methods allow the examination of these events in intact tissue in real time, enabling a greater understanding of cellular interactions and their interdependence.

The field of fluorescence microscopy has seen the principal advances. The development of new microscopic methods, coupled with the use of fluorescent proteins, new fluorescent dye technologies, highly sensitive detectors, and inexpensive powerful computers, have made this technique widely used in biomedical research. This review outlines the principles, potential application, and implementation of contemporary fluorescent imaging techniques in living cell systems.

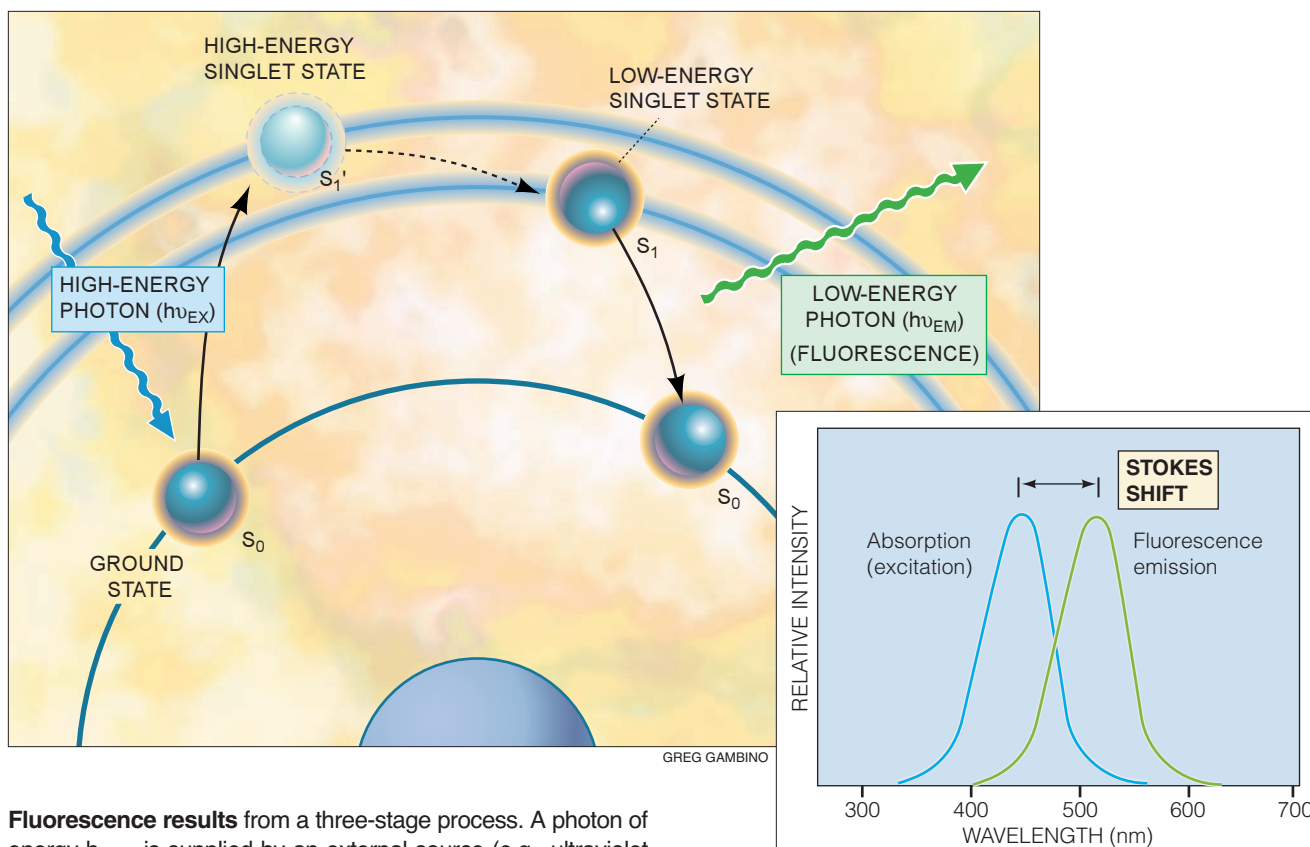
Electron Excitation and Relaxation Produce Fluorescence

Fluorescence microscopy depends on the inherent ability of a fluorescent molecule (fluorophore) to be excited by, and to emit light of, a specific and known wavelength. Generally, the excitation wavelength is shorter (more blue) than the emission wavelength.

If light is considered as photonic packets rather than electromagnetic waves, the process of fluorescence excitation can be considered a quantum event. Incident photons of a specific wavelength excite electrons within the fluorophore, generally in double bonds or pi orbitals, from the ground state to a defined higher-energy singlet state. The energy- or wavelength-selective nature of this transition occurs because electrons can only move between defined energy states with specific electrical, vibrational, and rotational components.

The time of electron excitation is short, in the femtosecond range. However, while the electron is in this excited state, it loses energy via internal conversion. Subsequently, the electron relaxes back to the ground state and releases a lower-energy or longer wavelength photon.

The difference in wavelength between the excitation and emission light is known as the *Stokes shift* and is the essential principle in all fluorescence measurements. The Stokes shift allows emission pho-



Fluorescence results from a three-stage process. A photon of energy $h\nu_{EX}$ is supplied by an external source (e.g., ultraviolet lamp or laser) and is absorbed by the fluorophore, exciting an electron to jump to a higher energy level (singlet state S_1'). The excited state exists for a short time (typically 1–10 nsec) and the energy of S_1' is partially dissipated (to S_1). A lower-energy photon $h\nu_{EM}$ is emitted, producing the fluorescence, as the fluorophore returns to its ground state S_0 .

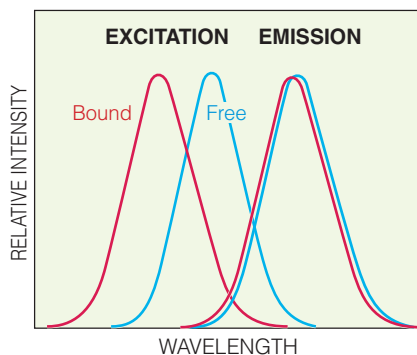
Stokes shift refers to the loss of vibrational energy when electrons relax from the excited state back to the ground state. As a result of this energy loss, the emission spectra of an excited fluorophore is shifted to longer wavelengths when compared to the excitation spectrum.

tons to be differentiated from excitation photons and the background of nonfluorescing molecules.

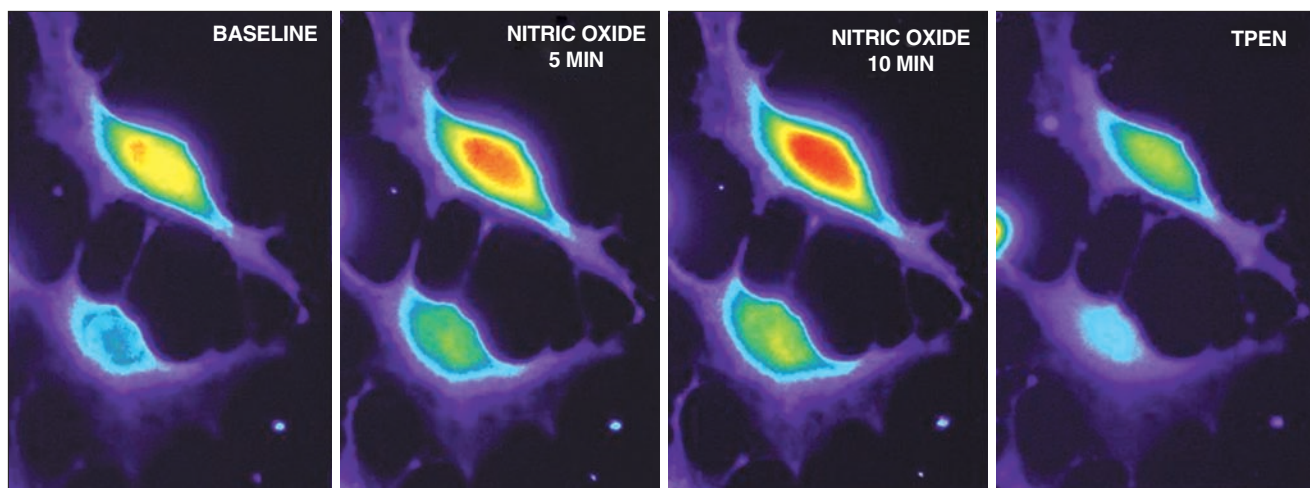
It also should be mentioned that dipole changes affecting the ionic charge balance within the molecule may occur, causing a shift in excitation or emission wavelengths. Generally, such changes are due to changes in the solvent characteristics (more or less polar), including increases in specific ionic species (e.g., Ca^{2+}). These wavelength changes may be measured and are the principle behind ratiometric dyes, such as the calcium indicator Fura-2, which changes excitation maxima on binding free calcium. With appropriate controls (chelating all calcium or flooding the system with calcium in the case of Fura-2), it is possible to make highly accurate measurements of ionic concentrations.

Instrumentation for Live Cell Imaging Controls Phototoxicity

Traditional microscopic tools, as with biochemical and/or molecular biology approaches, provide information as isolated static snapshots. To generate temporally useful data with these systems, it is necessary to perform multiple experiments at



Ratiometric indicator compounds have a shift in either their excitation or emission spectra on binding to ions. The excitation spectrum of ratiometric dyes such as the calcium dye Fura-2 changes according to the free Ca^{2+} concentration. The Ca^{2+} concentration is measured as the ratio between two fluorescence intensity values that are taken at two wavelengths.



Ion-sensitive fluorophores allow monitoring of intracellular parameters. In cultured mouse pulmonary artery endothelial cells, NO is shown to increase labile zinc. Endothelial cells were loaded with the zinc-sensitive fluorophore FluoZin-3 (Molecular Probes, Inc., Eugene, OR) and imaged at 37 °C using epifluorescent microscopy. The pseudocolored images show the increase in fluorescence over time in response to administration of the NO donor S-nitroso-cysteine ethyl ester and the rapid reversal of NO-induced increases in labile zinc by administration of the zinc chelator *N,N,N',N'*-tetrakis-(2-pyridylmethyl)-ethylenediamine (TPEN).

each time point, collect the data, and then use statistical analysis to build an overall picture of events occurring over time.

However, live cell imaging provides real-time images of cellular and subcellular events using multiple molecular reporters over extended periods of time and in three dimensions. To achieve live cell imaging, new tools have been developed that allow multiparametric analysis while maintaining the functional viability of the cells. Most laboratories use completely automated systems built around standard epifluorescence microscopes, as opposed to laser scanning confocal microscopes, and equipped with temperature-controlled stage inserts to maintain 37 °C (which preserves the physiologic environment).

The primary advantage of the epifluorescence scope is the substantially lower level of illumination required for imaging. Laser scanning confocal microscopy uses laser light to scan cells in a point-by-point fashion. Generally, the amount of light used in this technique is high (particularly by inexperienced users) and leads to quenching or photodestruction of the fluorophore and phototoxicity to the cell.

This process may occur in part due to excitation photons interacting with cellular components (e.g., lysosomes or other granular light-absorbing structures) to generate

heat. More commonly, excited fluorophores fail to transition back to ground state efficiently with the concomitant release of the lower-energy photon.

Minimizing the frequency of these events is a major issue in live cell microscopy and entails a careful and judicious selection of fluorophore. Nevertheless, management of the excitation light budget, regardless of the fluorophore, is a critical issue in live cell imaging and can be enhanced by the use of automated shutters and filter wheels or monochromators, which enable rapid switching of excitation wavelengths.

Various Fluorescent Probes Label Different Subcellular Structures

While enormous amounts of information have been gleaned from simple cell migration or tracking experiments, most live cell experiments typically combine transmitted light imaging with a fluorescent reporter tagged to either a subcellular structure (e.g., an organelle) or a cellular molecule (e.g., protein or nucleic acid).

The labeling of subcellular organelles can be achieved easily with membrane-permeant, organelle-specific dyes. A host of dyes is commercially available, with a variety of excitation and emission spectra that can label the plasma membrane, nucleus, mitochondria, and

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other organelles. Functional dyes also allow the fluorescent monitoring of intracellular parameters, such as pH, Ca^{2+} and other ion levels, and reactive oxygen and nitrogen species.

One example of the use of insensitive fluorophores in live cell imaging is provided in the figure, showing nitric oxide (NO)-induced, time-dependent increases in labile zinc in living primary cultures of mouse pulmonary artery smooth muscle cells, detected using the zinc-sensitive indicator FluoZin3 (Molecular Probes, Inc.). The specificity of this probe for zinc is shown by the decrease in fluorescence below baseline following application of the zinc chelator TPEN.

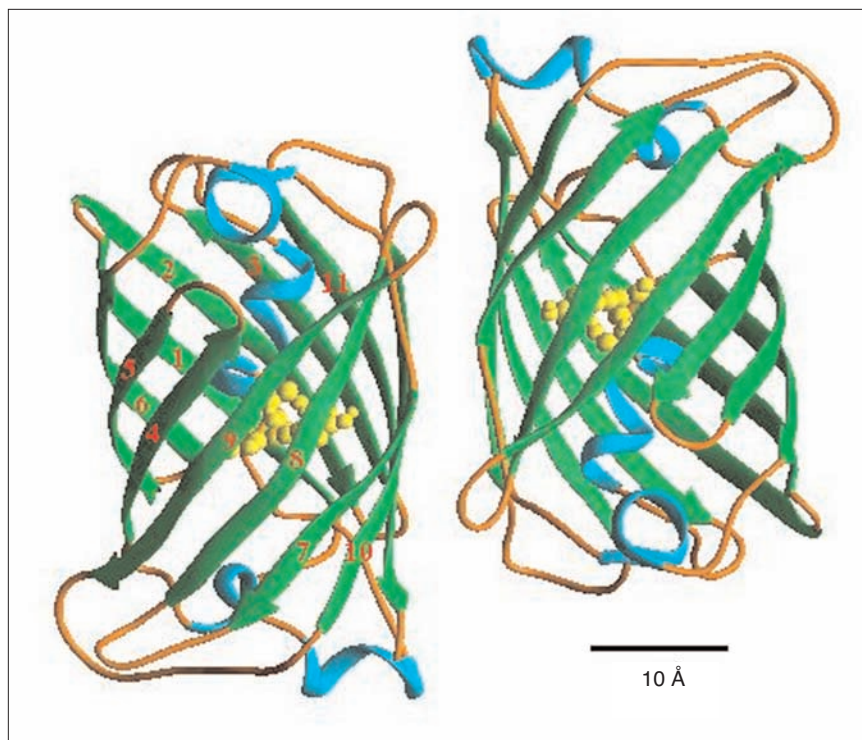
The labeling of individual molecules can be achieved with commercially available kits that react, for example, with free amine groups. Such kits can be used to label proteins of interest directly or, alternatively, to label primary antibodies targeting a molecule of interest.

This technique is very useful when directly examining cell-surface proteins or protein-protein interactions at the cell surface. For example, we have used antibodies conjugated directly to cell-surface receptors, as well as conjugated ligand, to study receptor internalization. However, the use of fluorescently conjugated molecules to study intracellular events typically requires the use of invasive methods, such as microinjection or scrape loading.

Genetically Encoded Fluorescent Probes Report Protein Behavior

Genetically encoded fluorescent probes are likely the most important development in optical imaging in the last decade. Most of these probes are based on green fluorescent protein (GFP). GFP is a spontaneously fluorescent protein originally isolated from the Pacific jellyfish, *Aequorea victoria*.

The fluorophore in GFP is deriv-



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ed from three consecutive amino acids: Ser65, Tyr66, and Gly67. Following synthesis of GFP, these three amino acids undergo autocatalytic cyclization and oxidation to form the functional fluorophore. The fluorophore contains a series of conjugated double bonds that provide the fluorescent properties of GFP.

Several aspects of the GFP molecule make it particularly attractive for use in live cell microscopy.

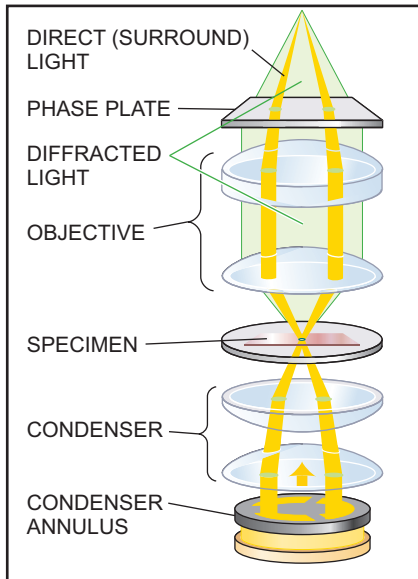
- First, it is simple to make constructs that contain GFP fused to the genetic sequence of a protein being studied. GFP consists of only 238 amino acids (29 kD) and can be fused to a variety of cellular proteins at either the COOH or NH_2 termini, without affecting the protein's native function or cellular localization.

- Second, GFP is highly stable, allowing its use in living cells as well as in fixed or frozen cells and tissues.

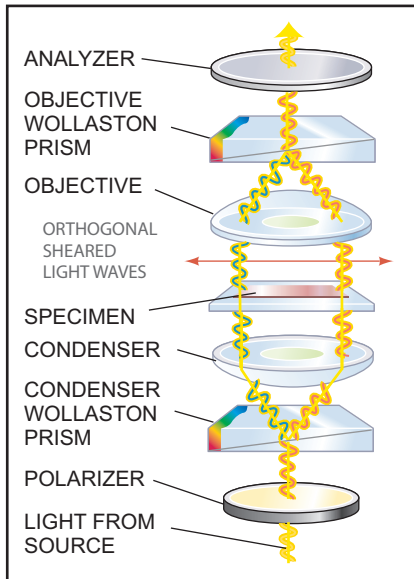
- Finally, mutations in the three amino acids comprising the fluorophore have produced GFP variants with differing spectral characteristics, thus increasing the number of potential uses for GFP.

Green fluorescent protein (GFP), shown in a stereoview of its three-dimensional structure, consists of 11 β -strands (green ribbons) forming a hollow cylinder through which is threaded a helix bearing the chromophore (yellow).

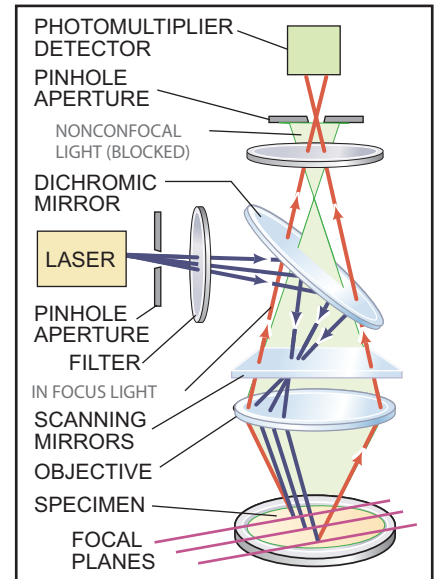
PHASE CONTRAST



DIC



CONFOCAL



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Phase contrast microscopy depends on two specialized phase plates. Light waves passing through the condenser annulus illuminate the specimen and either pass undeviated or are diffracted in phase by the cellular structures. The phase plate in the objective then collects the undeviated and diffracted light to form the final image.

Differential interference microscopy uses a Wollaston prism to create parallel wavefronts, which are differentially retarded by structures of different refractive indices in the specimen. When the wavefronts are recombined, destructive or constructive interference produces the final image.

Confocal microscopy uses a pinhole to focus the excitation light (blue rays) onto a single point in the specimen. A second pinhole aperture is placed in front of the detector, at a position that is confocal with the illuminating pinhole. This second pinhole prevents light from regions outside the plane of focus (nonconfocal) from reaching the detector. A two-dimensional image of the precise plane of focus is created by scanning the beam of light across the specimen using two oscillating mirrors placed between the dichromatic mirror and the objective lens in such a way that the illuminating spotlight and confocal pinhole at the detector remain strictly in register.

For example, a double mutation of Phe64→Leu and Ser65→Thr yields enhanced GFP (EGFP), the most commonly used GFP variant. EGFP exhibits a red-shifted excitation maximum from 395 nm (wt) to 488 nm, allowing for the use of standard fluorescein isothiocyanate (FITC) optics, and it is more photostable and brighter than the wild-type GFP.

GFP-based reporters have proved to be invaluable tools to study complex biochemical processes in real time, including the tracking and quantification of individual or multiple proteins and the monitoring of protein–protein interactions. In addition, GFP-based indicators have been used as photo-modulatable proteins to highlight and follow the fate of protein populations within a cell and as biosensors to describe biological events and signals.

Transmitted Light Microscopy Enhances Capabilities of Fluorescence Techniques

In addition to fluorescent methods, transmitted light illumination is essential to any live cell imaging system. At a minimum, light microscopy is used only to validate the specimen, but because live cell imaging does not allow contrast of cells and tissues by traditional staining methods, other noncolorimetric brightfield-contrasting methods must be used. However, these different methods may affect the efficiency of fluorescent light excitation and detection of the emitted fluorescent signal.

Phase contrast microscopy depends on the presence of a pair of phase plates: one in the condenser and the other in the objective. As fluorescence imaging uses the objective for transmission of both

the exciting light and emitted signal, this phase plate can interfere with the signal recovered (up to 30% of the available signal may be lost). For this reason, this contrast method is generally avoided for any instrument trying to detect a low-level fluorescent signal. Phase contrast is still useful when imaging cytoskeletal components in transmitted light, as these cellular features generally provide only minimal contrast on differential interference contrast.

Differential interference contrast (DIC) is the preferred method for high-resolution transmitted light imaging in live cell microscopy. DIC uses differences in the refractive index of closely apposed structures (e.g., at the edge of a cell, the lipid bilayer is adjacent to the aqueous media) to generate a high-resolution image. This technique allows exquisite resolution of cellular motion.

DIC depends on polarized light that is then effectively split into plane-parallel bundles using a birefringent material (e.g., calcite) within the microscope; this is known as a *Wollaston prism*. The light then passes through the sample, is recombined by a second Wollaston prism, and lastly passes through a final polarizing filter, the analyzer.

Although the Wollaston prism has little effect on light transmission, the polarizing analyzer filter can confound fluorescence signals (for both excitation and emission) considerably. However, it is possible to remove the DIC analyzer during fluorescence detection to avoid interference when it is not being used.

Motility assays are the most basic use of DIC imaging technology, used to visualize such things as sperm motility and cell migration.

Perhaps the most significant aspect of DIC imaging and motility measurements is that transmitted light, rather than fluorescent emission, is used as the collected signal. This illumination is low intensity, causing negligible phototoxicity to

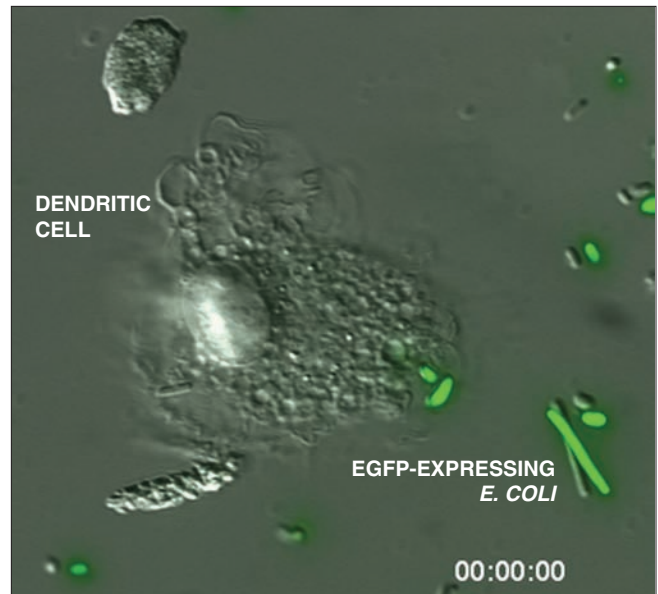
the cell. In fact, microscopes for simple cell motility assays can be quite simple, as the images may be collected at low light levels with no shutters or automation needed for fluorescent emission experiments.

In one recent motility experiment, we used DIC combined with GFP as a fluorescent label attached to bacterial proteins to follow dendritic cell uptake of bacteria. High-resolution live cell imaging of EGFP-expressing *Escherichia coli* was used to obtain a detailed picture of particle uptake by dendritic cells derived from human peripheral blood monocytes.

As the DIC techniques showed, immature dendritic cells undergo pronounced morphologic changes within minutes on exposure to nonopsonized *E. coli*. The dendritic cells develop extensive membrane veils that efficiently capture multiple bacteria. Internalization does not occur in the veils, but instead, bacteria are transported to the central region of the cell, where they sink directly into the plasma membrane (see video 1).

Combining Contrast and GFP-Based Imaging Yields Advanced Techniques

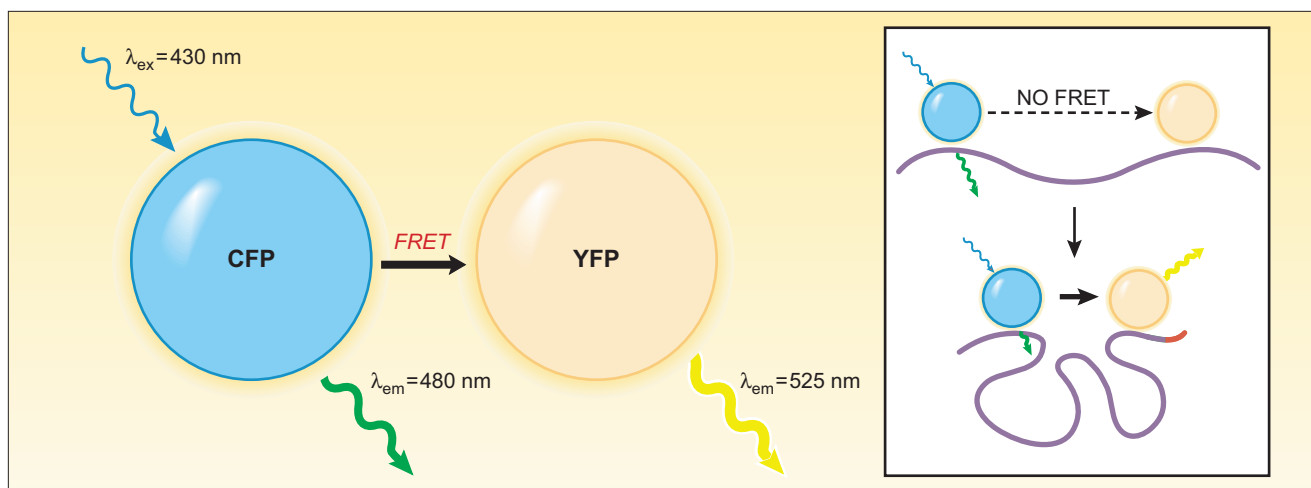
The advent of GFP cloning technologies has provided a means of making specific proteins visible in living cells, and one of the most common applications of GFP in live cell microscopy has been as a label for specific proteins, allowing the localization and trafficking of the protein in vitro. A number of



SALTER RD, ET AL: J LEUKOCYTE BIOL 75:240-243, FEB 2004.

DIC techniques show uptake of live EGFP-expressing *E. coli* by human monocyte-derived dendritic cells. Within minutes of opsonization, the dendritic cells develop extensive morphologic veils that capture other bacteria.

🔗 See **video 1** associated with this paper at www.sciandmed.com/sm/.



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Fluorescence resonance energy transfer (FRET). The donor fluorophore (*cyan*) is excited by light of a specific wavelength. The excited state energy is transferred nonradiatively to the acceptor fluorophore (*yellow*) which then emits light at its characteristic wavelength.

In one FRET-based application (*inset*), donor and acceptor GFP fluorophores are incorporated at opposite ends of a protein. As protein conformation changes, the relative positions of the two fluorophores are altered, causing a detectable change in emissions of both the donor and acceptor.

sophisticated methods have been developed to study molecular behavior and interactions with high spatial and temporal resolution.

One of the earliest methods used to investigate steady-state protein dynamics was *fluorescence photobleaching*, in which a fluorophore is made nonfluorescent by exposure to high-intensity light. The bleached region serves as a reference mark to track labeled proteins. Derivations of photobleaching methods include fluorescence recovery after photobleaching (FRAP), fluorescence loss in photobleaching (FLIP), and fluorescence localization after photobleaching (FLAP).

Various other fluorescence-based technologies are useful for specific applications, but the two techniques used most commonly in our laboratories are fluorescence resonance energy transfer (FRET) and total internal reflection fluorescence (TIRF).

Fluorescence Resonance Energy Transfer (FRET)

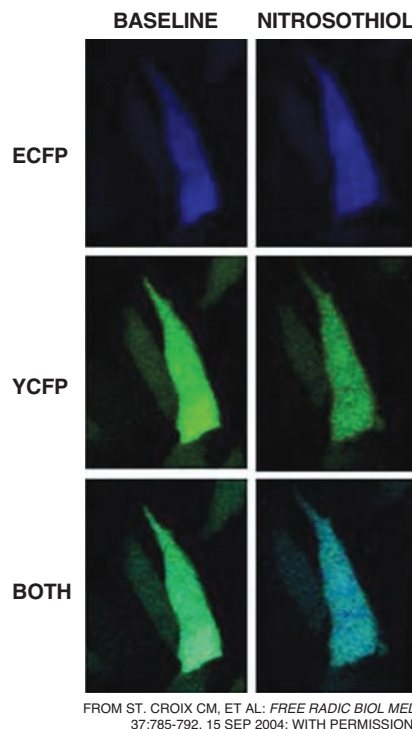
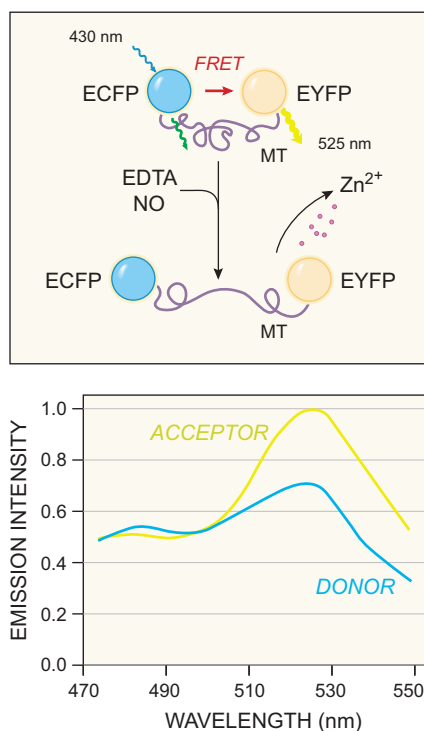
The development of GFP variants with differing absorbance and emission spectra has allowed for the simultaneous visualization of multiple tagged molecules within a cell. In particular, the pairing of the cyan (CFP) and yellow (YFP) fluorescent proteins has been crucial in advancing the use of FRET to study inter- and intramolecular interactions in living cells.

FRET is a quantum mechanical process whereby two fluorophores (i.e., a donor and an acceptor) that have appropriate spectral properties and are closely apposed (10-50 Å) transfer photon energy in a non-radiative fashion. The FRET effect decreases rapidly as the distance between the donor and acceptor fluorophores increases.

Two general strategies have evolved using GFP variants in FRET-based applications. The first involves the incorporation of GFP donor and acceptor fluorophores at opposite ends of a conformationally active protein. Alterations in protein conformation change the relative positions of the two fluorophores, and this change is reflected in alterations in FRET.

The second strategy involves the encoding of GFP donor and acceptor fluorophores on two distinct proteins. In this case, FRET between the two fluorophores provides evidence for a physical interaction between the two proteins. This method has been used widely in living cells to support *in vitro* data provided by immunoprecipitation and other biochemical techniques.

Verification of the occurrence of FRET, in both inter- and intramolecular applications, is obtained by using selective acceptor photobleaching to generate an increase in donor fluorescence. A second control for demonstrating intermolecular FRET is to show that biologically inert mutants of the



FRET-based detection of metallothionein-NO interaction. A FRET-MT reporter was constructed of human type IIa metallothionein (MT), flanked by ECFP and EYFP molecules. MT unfolding, induced by metal chelators or nitric oxide (NO), results in metal (zinc) release and consequent changes in FRET.

In this experiment, pulmonary endothelial cells were infected with an adenoviral vector encoding the fluorescent FRET-MT reporter molecule and imaged 24 hours later. FRET was detected in real time, using full spectral confocal imaging. The graph shows the separation of the two emitted signals (cyan and yellow) following spectral unmixing based on individual calibration spectra for each protein.

The spectral report (*far right*) shows an increase in the peak of the donor (cyan) and decrease in that of acceptor (yellow) following a 5-min exposure to an NO donor.

interacting proteins do not undergo FRET.

Methods for detecting FRET between a donor and an acceptor with overlapping excitation and emission spectra require the use of narrow detection bands and automatic switching of optical filters to differentiate between emissions, along with complex mathematical corrections to account for crosstalk between channels.

Recent advances in detector technology, however, now enable the resolution of fluorescent images, providing full spectral information for each voxel of the image without the need for switching optical filters. This method uses confocal imaging and is one situation in which confocal methods may be more appropriate than widefield imaging for live cell studies. With confocal methods, it is easy to selectively photobleach acceptor molecules, and by using calibration spectra, the cross-talk between overlapping cyan and yellow emissions can be separated. The use of confocal methods allows the detection of small, but potentially biologically meaningful, changes in FRET that are common with gen-

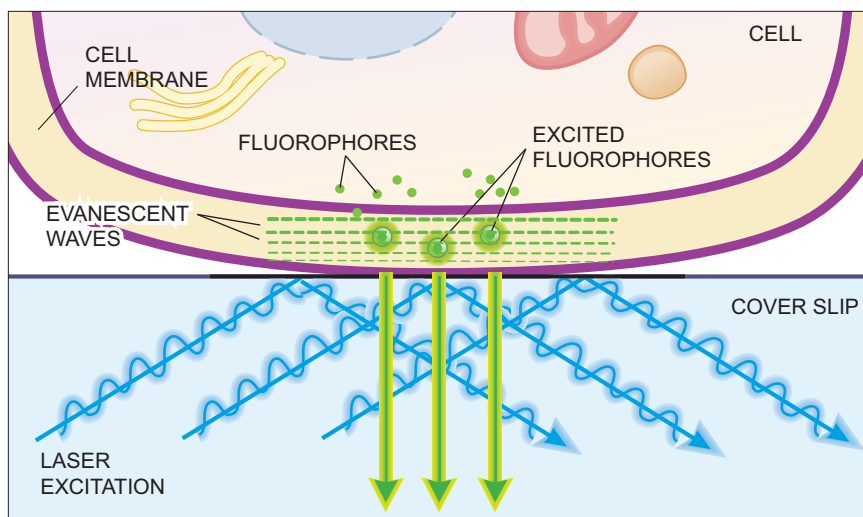
etically encoded reporters and that are extremely difficult to resolve reliably using more traditional methods based on bandpass filters.

We used confocal-based spectral imaging to examine the role of metallothioneins (MT) in nitric oxide signaling. Previous data had suggested that metal ion homeostasis is regulated in part by a link between MT and cellular redox status via the interaction of MT with the free radical NO.

We used a chimera reporter fusion protein consisting of MT sandwiched between enhanced CFP (ECFP) and enhanced YFP (EYFP) molecules, and FRET between ECFP and EYFP was used to monitor conformational alterations in the core MT protein. Alterations in the FRET-MT signal, indicative of metal release, were produced by exposing live, FRET-MT-expressing pulmonary endothelial cells to NO donors or agents that activate intracellular NO synthesis.

The GFP-FRET method also has been used to study a variety of other intracellular signaling pathways using reporter molecules for calcium, cGMP, cAMP, and tyro-

Total internal reflection fluorescence microscopy (TIRF). When an excitation light beam traveling at a low angle through a dense medium (e.g., glass coverslip) reaches an interface with less dense medium (e.g., aqueous solution, cell surface), it causes all of the light to be reflected at the cell surface rather than passing through the cell. The reflection creates a thin evanescent field 100 to 200 nm deep. Fluorophores near the cell surface or interface are excited by the evanescent wave, but fluorophores farther away are not. This effect produces a high-contrast image of events occurring at or near the cell surface.



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sine kinase activity, among others.

A potentially powerful way to detect FRET is by combining FRET with a second advanced imaging modality known as *fluorescence lifetime imaging microscopy* (FLIM). FLIM measures the extent to which a donor's fluorescence lifetime is shortened by FRET events, using a pulsed laser to excite the probes and a gated camera to detect the fluorescence of the donor.

The use of FLIM to measure FRET efficiency is particularly useful when there is significant overlap of the donor fluorescence and the acceptor emission wavelengths or when the donor is prone to photobleaching. Combining FRET and FLIM technologies, however, requires highly sophisticated equipment and a great deal of technical expertise.

Total Internal Reflection Fluorescence Microscopy (TIRF)

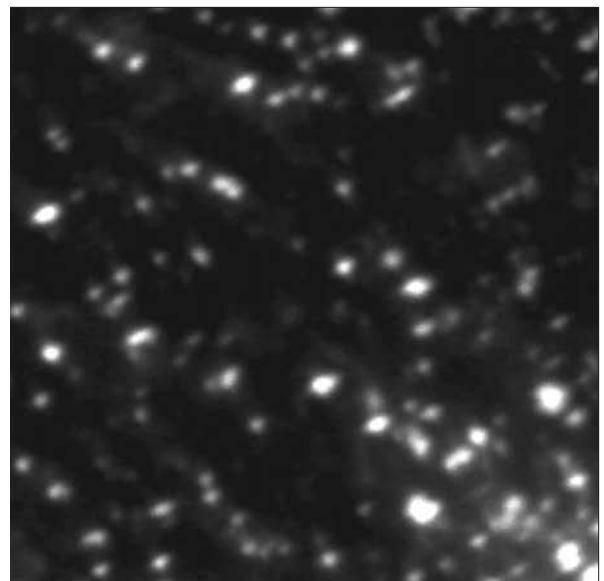
The physical phenomenon of total internal reflection relies on the refraction (bending) of light as it encounters the edge between two media with different refractive indexes, resulting in the confinement of some or all of the light to the higher index medium. In TIRF, light from laser sources is directed at a specimen on a glass coverslip in such a way as to create an evanescent wave propagating parallel to the interface between the cover-

slip and the aqueous medium due to nearfield effects. The intensity of such a wave falls off exponentially with distance from the coverslip, resulting in excitation of only the fluorophore at the portion of the cell nearest the coverslip (thereby establishing optical slices of <100 nm).

As no photons are used for illumination, this method not only provides emission from a limited, defined focal volume but also offers the highest signal-to-noise image currently possible in light microscopy. In addition, with careful control of light intensity, photobleaching and phototoxicity are reduced.

The basic principles of TIRF were described in the early 1980s, but the recent development of improved objectives, sensitive detectors, and genetically encoded GFP-based reporters has led to a resurgence of interest in its application. TIRF is currently used, either by itself or in combination with other optical approaches, to study various physiologic events associated with membrane trafficking, vesicular transport, secretion, and exocytosis. The growing availability of multicolored TIRF systems will greatly increase the number and complexity of molecular questions to be resolved with this technology.

We used simultaneous TIRF to characterize the dynamics and interactions of components of the



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endocytotic machinery. In eukaryotic cells, a clathrin coat forms on membranes as a polygonal lattice, which progressively curves to incorporate membrane, cargo, and extracellular fluid into a transport vesicle. The kinetics of clathrin coat assembly and the mechanics of adaptor protein interaction are not well understood.

We used TIRF microscopy to extend the time-resolved study of clathrin structures by analyzing the temporal co-localization of EGFP-clathrin with fluorescently labeled, compartment-specific adaptor proteins and/or cargo molecules. We showed that most clathrin structures are relatively static, moving vertically in and out of the evanescent field, but with little lateral motion. A small minority of clathrin-positive puncta, however, was distinct from plasma membrane clathrin lattices; these were highly motile and associated with endocytosed cargo molecules, revealing an endosomal population of clathrin structures (see video 2).

Intravital Imaging Permits Study of Intact Living Tissue

Some established experimental models have been adapted to allow for the direct fluorescent imaging of intracellular events in intact tissues. In particular, the ability to

isolate and perfuse the circulatory systems of either the heart or lungs of rodents has enabled researchers to use a variety of cell-permanent fluorescent probes, including those sensitive to ion flux, redox status, and pH, to study the integrative nature of signaling events at both the subcellular and intercellular levels.

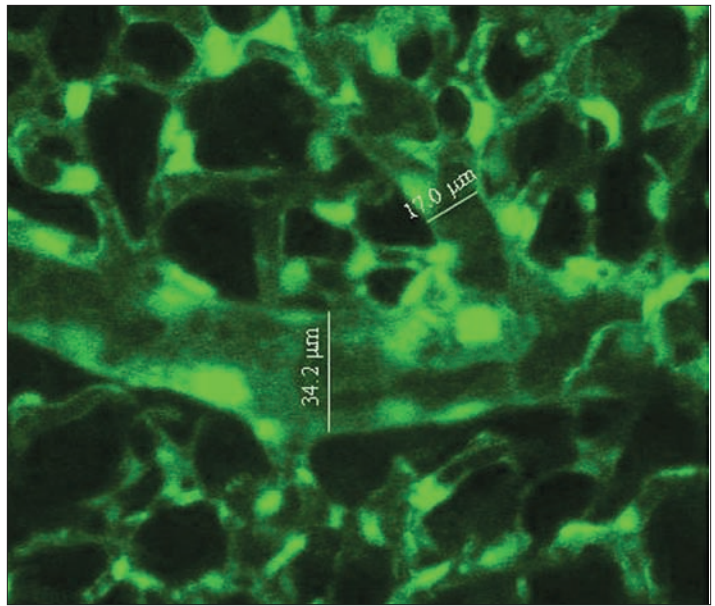
In addition, fluorescent imaging techniques have been used to study the vascularization of organs and their regulation of blood flow, including measurement of the distribution and movement of fluorescent beads or indicators present in the perfusate. In fact, *in situ* imaging has always been at the forefront of contemporary pulmonary research, ever since video-microscopy was used to determine pulmonary capillary transit time by measuring the time required for fluorescent dye to pass from an arteriole to a venule on the surface of an intact dog lung.

More recently, both epifluorescent and confocal microscopy techniques have been used to image isolated, perfused rat lung. These studies were intended to: 1) observe intercellular calcium signaling at the alveolar-capillary junction; 2) study the role of reactive oxygen species (ROS) in ischemia/reperfusion injury in lung tissue; and 3) measure transalveolar transport of

Clathrin dynamics in living cells were visualized with TIRF microscopy. The left panel shows a single human fibroblast expressing GFP-labeled clathrin light chain α (GFP-LC α). Motion within the boxed region is shown by color coding of GFP-LC α in the right panel, with white representing stationary objects.

🔗 See **video 2** associated with this paper at www.sciandmed.com/smf/.

Laser scanning confocal microscopy of the sub-pleural vasculature of a perfused lung isolated from a Tie2-GFP mouse. The TIE2-GFP mouse (Jackson Labs.) expresses GFP under the direction of the endothelial-specific receptor tyrosine kinase (Tie2) promoter and thus can be used to visualize pulmonary vessels. Using sequential XYZ-sections (512 X 512 pixels, at Nyquist axial frequency 250 nm) which include the entire vessel being studied, it was possible to reconstruct the vessel and generate real measures of cross-sectional area and maximal diameter using commercially available software. The white bars mark the diameters of small pulmonary vessels.



water, protons, and solutes by adding membrane-impermeant fluorophores to the airspace fluid.

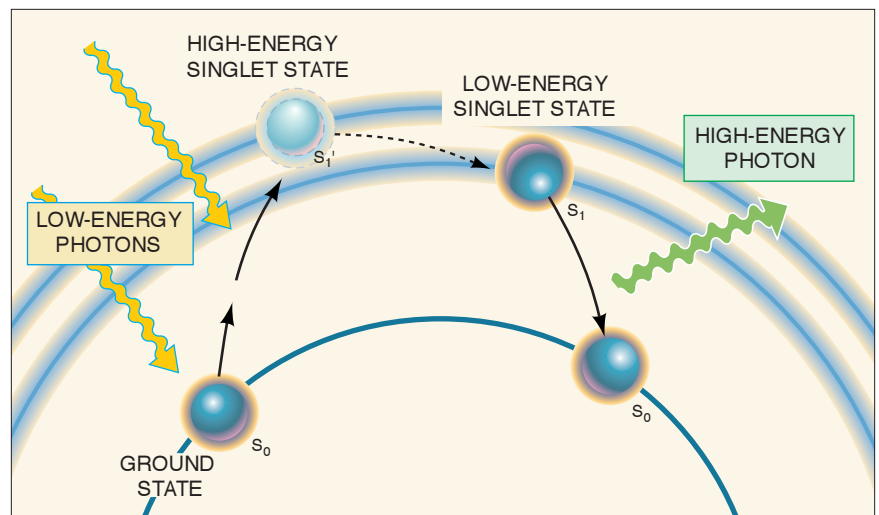
The use of specific promoters and targeting signals to introduce GFP-tagged proteins into the intact organism and to direct the proteins to specific tissue regions, cell types, and subcellular compartments has provided spatiotemporal information in complex tissues, thereby placing it in a physiologic context.

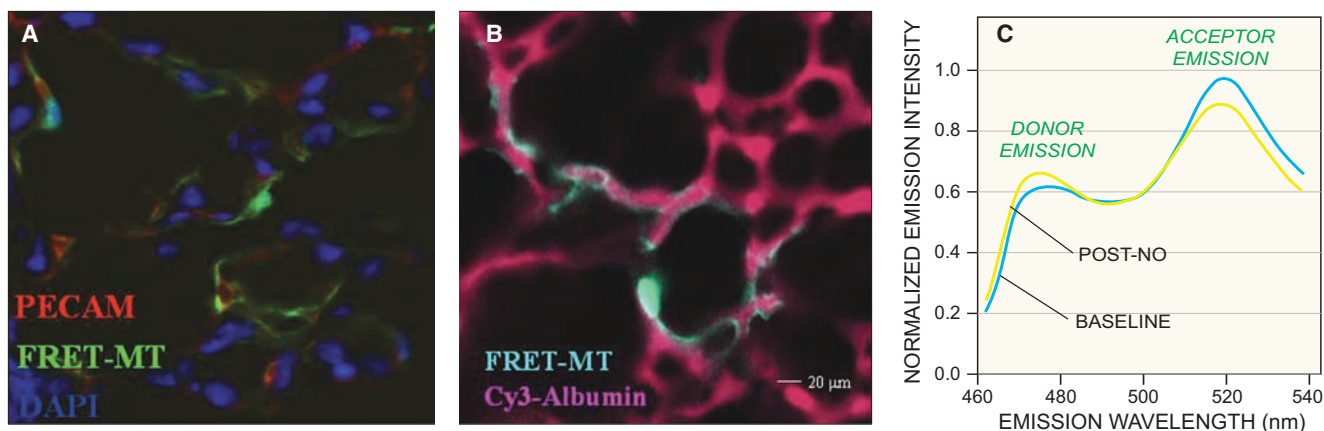
The utility of this approach was demonstrated in an experiment using confocal-based imaging to study surface vasculature in the isolated perfused lung of a TIE2-GFP mouse. This mouse expresses

GFP under the direction of the endothelial-specific receptor tyrosine kinase (Tie2) promoter. Using sequential XYZ-sections, the three-dimensional anatomy of the vasculature was reconstructed, and real measures of cross-sectional area and maximal diameter were generated. This application is potentially useful for measuring vascular reactivity in response to either inhaled or circulating agonists.

Significant advances in gene transfer technologies in recent years also permit the introduction of genetically encoded probes into intact animals to study physiologic signaling events in their appropriate microenvironment. For exam-

Multiphoton laser scanning microscopy of deep tissue uses two long-wavelength (near-IR) photons to generate the same electronic transition as a single shorter wavelength photon.





FRET detection in the isolated perfused mouse lung. Expression of the FRET-MT reporter in pulmonary endothelium was achieved via injection of DOTAP:cholesterol liposomes followed by adenovirus containing cDNA for FRET-MT.

A, Confocal imaging of a section of murine lung tissue 48 hrs after injection shows co-localization of staining for endothelial-specific platelet/endothelial cell adhesion molecule (PECAM, red) and FRET-MT expression (yellow), suggesting expression is predominantly endothelial. DAPI (4',6-diamidino-2-phenylindole) binds to dsDNA and labels nuclei of all cell types.

B, Confocal imaging of the isolated mouse lung, perfused with labeled albumin (magenta), shows the vascular network, whereas FRET-MT expression (yellow) is largely confined to the microvasculature.

C, Spectral laser scanning confocal imaging of the subpleural vasculature of the isolated perfused mouse lung shows that the addition of the NO donor DETA-nonoate to the perfusate causes an increase in the peak emission intensity of the donor (cyan ~485 nm) and a decrease in that of the acceptor (yellow 525 nm), consistent with conformational changes in FRET-MT.

ple, endothelial expression of the FRET-MT reporter in mouse lung was achieved using injections of DOTAP:cholesterol liposomes followed by adenovirus containing cDNA for FRET-MT. Spectral laser scanning confocal imaging of the subpleural vasculature of the isolated perfused mouse lung confirmed that NO donors caused conformational changes in the FRET-MT reporter in the intact pulmonary endothelium, as previously shown in live cultures of endothelial cells described earlier.

The imaging of intact tissue can provide information about the spatial distribution of biological phenomenon over time within a particular organ, provided that the chosen technology has the appropriate depth penetration and the ability to perform computational three-dimensional reconstruction of images after acquisition. As with live cell microscopy, maintaining the viability of the tissue is also a concern during serial imaging.

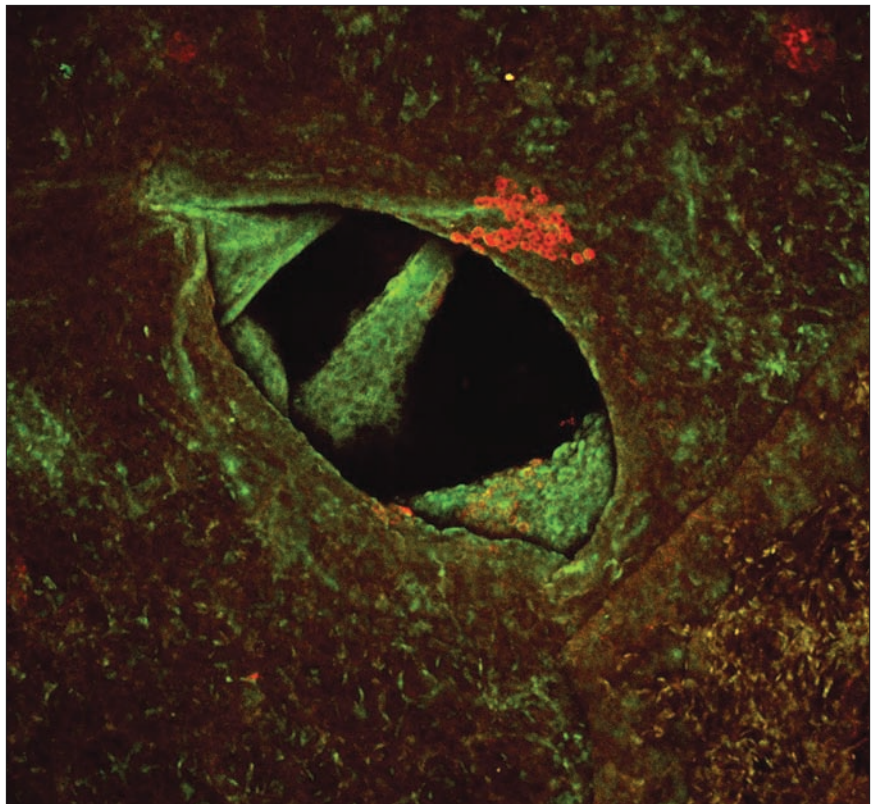
Confocal approaches are generally limited by the scattering and absorbance of light entering the

tissue. This limits the localization of excitation, while scattering of the emitted photons by the tissue causes much of the emitted signal to be lost. To successfully image deep within light-scattering material, it is necessary to use a long-wavelength illumination system together with a detection path that is minimally affected by scattering.

Although some of the limitations of short-wavelength scattering (e.g., for blue light at 488 nm) by living tissue can be overcome by operating in the near-infrared region of the spectrum, this approach poses its own limitations. Currently only a few near-infrared-excitable fluorescent dyes are available for application in cellular imaging, and although the longer emission wavelengths are less prone to scattering, the detection of emitted photons is still subject to projection through a detection pinhole to reject out-of-focus signal. Thus, multiphoton imaging has become the method of choice for these types of studies.

Multiphoton laser scanning microscopy (MPLSM) relies on the simultaneous absorption of two

Multiphoton laser scanning microscopy shows a high-resolution three-dimensional image of the distal conducting airways of the lung. The red fluorescence marks cells that are immunoreactive for Clara cell secretory product.

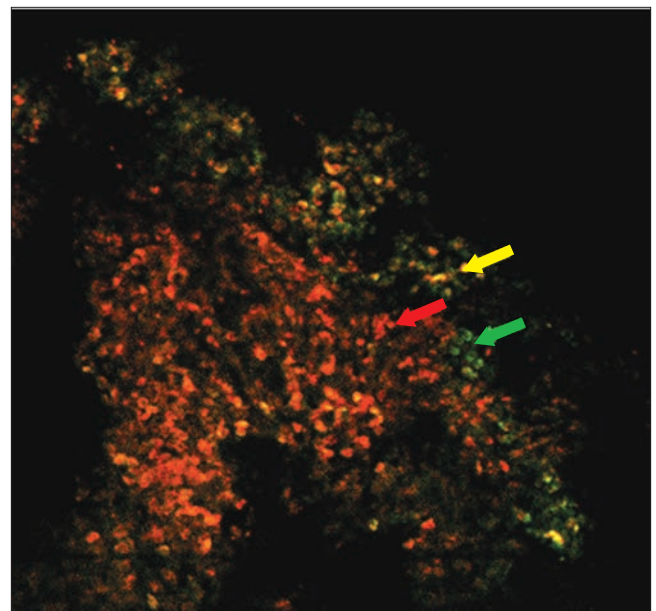


long-wavelength (near-infrared) photons to generate an electron transition within a fluorophore equivalent to that produced by a single high-energy photon. The diffraction-limited focusing and

pulsed nature of the lasers used result in an excitation event only at the plane of focus of the illuminating lens, avoiding the need for a pinhole aperture. Also, there is less phototoxicity.



In vivo imaging of insulin-Timer islets transplanted under the renal capsule of a mouse and accessed via a transdermal window affixed to the flank. On multiphoton laser scanning microscopy (MPLSM, *right*), the timer protein changes from green to red over the first 24 hours after synthesis and therefore can be used to monitor the rate of insulin production in images taken over an extended time period. At the single time



point shown, pockets of newly synthesized insulin (*green arrow*) and areas of more mature protein (*yellow to red*) are seen within the same field of islets.

Intravital Imaging Enables In Situ Visualization in Intact Living Animals

The ultimate goal of biomedical research is to identify, follow, and quantify biological processes at the cellular and subcellular level in the intact organism. One means of achieving this goal has been with the development of chronic animal models incorporating transparent windows, enabling the visualization of temporal physiologic events using intravital microscopy. These techniques, in combination with surgical implantation of GFP-transduced cancer cell lines into metastatic rodent models, have been particularly useful in the study of tumor vascular function and tumor-host interactions.

As in deep tissue imaging, the advantages of multiphoton laser scanning microscopy (MPLSM) for acquiring three-dimensional data, while limiting light scattering and phototoxicity, are particularly suited for high-resolution in situ imaging. The value of combining advances in transgenic technologies with increasingly sophisticated imaging modalities, to address physiologically relevant questions, was illustrated by the development of a transgenic mouse expressing proinsulin II tagged with the fluorescent reporter protein, Timer, to investigate insulin secretion.

Timer is a mutated version of a red fluorescent protein from the *Anthozoa* coral. The mutations permit its fluorescence to change from

green, when it is first synthesized, to red over a period of about 24 hrs. The age of the protein tagged with Timer can be determined by the observed ratio of green to red fluorescence. Pancreatic β -cells in these mice fluorescently reveal when they are producing insulin and how long it has been since they synthesized it.

Transplantation of Ins-C-Timer islets under the renal capsule has permitted the in vivo visualization, by MPLSM, of insulin synthesis through a body window device implanted over the transplant site. The window technique allows for multicolor imaging, over a period of several days, of both insulin-producing cells and T cells labeled with a fluorophore distinguishable from Timer. This model may prove useful in investigating the etiology of type 1 diabetes, as well as in monitoring the success of therapies based on allotransplantation of islet cells.

The increasing versatility of imaging probes, coupled with dramatic improvements in instrumentation and analysis tools, have provided important insights into myriad aspects of biological function at the subcellular, cellular, tissue, and organismic level. The primary challenges ahead involve the translation of these new tools and the knowledge base they have spawned in the understanding of biological responses to environmental variables.

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